

Name: _____

Date: _____

Assignment 1: Kinematics

Knowledge	Thinking/Inquiry	Application	Communication
3/4	4/4	4/4	4/4

INSTRUCTIONS

Answer all questions fully and ensure the following:

- Show all your work/steps including all calculations
- Write the main/general equation(s)
- Display proper notation for vectors (i.e. include arrows for vector variables)

Make sure to include the following in your final answer:

- Appropriate significant digits
- Proper units
- Direction (if the answer is vector)

KNOWLEDGE

1. A flight of stairs is 5 m high. If you were to run up and down the stairs 10 time.

- a) What is your total distance travelled?

$$5\text{m} \times 10 \times 2 = 100\text{m travelled}$$

[1 mark]

- b) Determine your total displacement.

up & down 10 times, still @ original position

$$\therefore \text{displacement} = 0\text{m [U]}$$

[1 mark]

2. A rabbit, initially hopping at 0.4 m/s, sees a fox and accelerates 1.4 m/s² for 2.5 s.

- a) What is the final velocity of the rabbit?

$$V_1 = 0.4\text{m/s} \quad V_2 = ?$$

$$a = 1.4\text{m/s}^2 \quad V_2 = V_1 + a\Delta t$$

$$\Delta t = 2.5\text{s}$$

$$V_2 = 0.4\text{m/s} + 1.4\text{m/s}^2 \cdot 2.5\text{s}$$

$$\therefore V_2 = 3.9\text{m/s} \text{ (Forward)}$$

[1 mark]

- b) Determine total distance the rabbit ran for 2.5 s.

$$V_1 = 0.4\text{m/s} \quad a = 1.4\text{m/s}^2$$

$$V_2 = 3.9\text{m/s} \quad \Delta t = ?$$

$$\Delta t = 2.5\text{s}$$

$$\Delta d = V_1 \Delta t + \frac{1}{2} a \Delta t^2$$

$$\Delta d = 0.4\text{m/s} \cdot 2.5\text{s} + \frac{1}{2} \cdot 1.4\text{m/s}^2 \cdot 2.5\text{s}^2$$

$$\Delta d = 5.375\text{m} + 0.4\text{m} = \Delta d$$

[1 mark]

THINKING / INQUIRY

3. How much time does it take for an F-22 fighter jet to accelerate from Mach 1 to Mach 2 at a rate of 40 m/s²?

Note: speed = (Mach number) X (speed of sound which is 332 m/s). Round the answer to one decimal place.

$$V_1 = \text{mach 1} = 1 \times 332\text{m/s} = 332\text{m/s}$$

$$V_2 = \text{mach 2} = 2 \times 332\text{m/s} = 664\text{m/s}$$

$$a = 40\text{m/s}^2$$

$$\Delta t = ?$$

$$V_2 = V_1 + a\Delta t$$

$$a\Delta t = V_2 - V_1$$

$$\Delta t = \frac{V_2 - V_1}{a}$$

$$\Delta t = \frac{664\text{m/s} - 332\text{m/s}}{40\text{m/s}^2}$$

$$\Delta t = 8.3$$

[2 marks]

4. A tennis ball makes contact with a tennis racquet and changes velocity from 14 m/s [E] to 24 m/s [W] in 0.20 s. Determine the acceleration of the tennis ball.

$$\vec{v}_1 = 14 \text{ m/s [E]}$$

$$\vec{v}_2 = 24 \text{ m/s [W]} = -24 \text{ m/s [E]}$$

$$\Delta t = 0.20 \text{ s}$$

$$\vec{a} = ?$$

$$\vec{a} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t}$$

$$\vec{a} = \frac{-24 \text{ m/s [E]} - 14 \text{ m/s [E]}}{0.20 \text{ s}}$$

$$\vec{a} = -190 \text{ m/s}^2 \text{ [E]}$$

[2 marks]

APPLICATION

5. Two friends see each other in a park. Initially, they are 30 m apart. The first friend starts walking toward the second friend at a constant speed of 0.2 m/s. At the same instant, the second friend accelerates uniformly from rest at a rate of 0.8 m/s^2 toward the first friend. How long before the two friends can fist bump each other.

let's be
first friend
travelling [E]

$$a_1 = 0$$

$$v_1 = ?$$

$$a_2 = 0.8$$

$$v_2 = 0.2$$

$$v_3 = 0.2 \text{ m/s [E]}$$

$$v_4 = 0.2 \text{ m/s [E]}$$

$$a_5 = 0.8 \text{ m/s}^2 \text{ [W]}$$

$$a_6 = 0.8 \text{ m/s}^2 \text{ [W]}$$

$$v_1 = \frac{a_1 \Delta t}{1}$$

$$a_2 = 0.8$$

$$a_3 = 0.8$$

$$a_4 = 0.8$$

$$a_5 = 0.8$$

$$a_6 = 0.8$$

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